



XRL XRL

Explainable Reinforcement Learning (XRL) for Autonomous Driving

Problem

Existing Demo of VLM

VLAM = VLM + A (DL /RL)

Planning to → Control [(Text based) Challenge]

Query: Red car is **turning** in which direction?

Response: The red car is turning to the right.

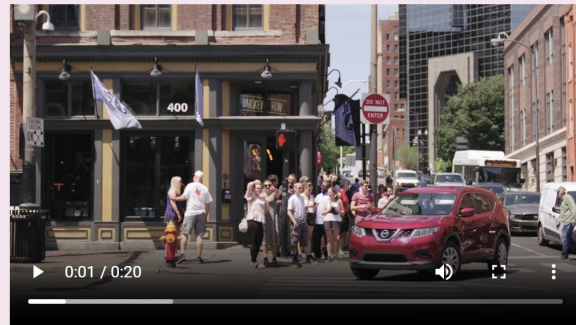
Asking a VLM about action

Action: Turning left or right.

Q: How to explain or interpret this action?



Red car is turning in which direction? Going left or right.



Running Vision Language Model (ViLA) on edge device

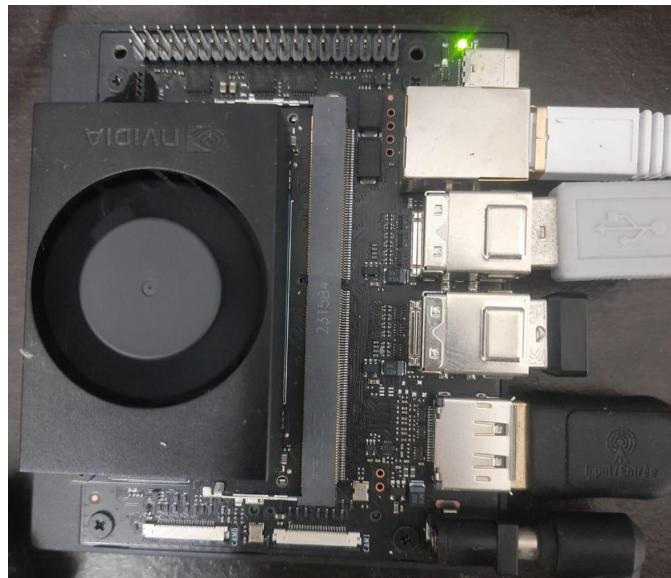
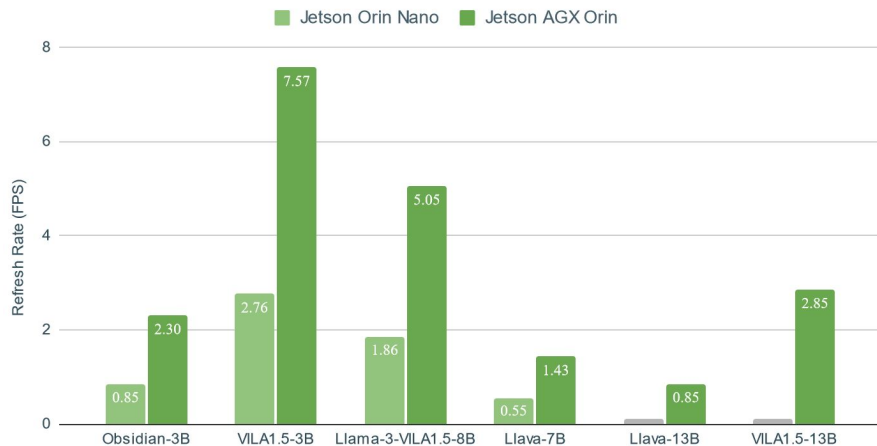
Optimized VLM for real time

Model running on **Jetson Orin (8GB)**

Decent (FPS)

Multimodal Streaming Rate

Vision Encoder + Projector + VLM



- Ref: [VILA: On Pre-training for Visual Language Models](#) (2024)

Testing on Images

Input Image

Prompt: **What does the road sign say?**

The road sign says “Hoover”



Output

```
>> PROMPT: 'what does the road sign say?'
```

```
The road sign says "Hoover."</s>
```

embed_time	0.000312776
input_tokens	28
output_tokens	12
prefill_time	0.0767153
prefill_rate	364.986
decode_time	0.475165
decode_rate	23.1499

Image

Environment state dynamics

White bus **parked** on the side of the road.

White bus **driving** down at **night**.



Two different **actions** explanations for same image.

Video (daytime)

Driving car with pedestrians

Original Video



Output Video



```
A red car is driving down a street with people walking on the sidewalk.  
18:13:49 | INFO | refresh rate: 0.46 FPS (2188.2 ms)  
A red car is driving down a street with people walking on the sidewalk.  
18:13:52 | INFO | refresh rate: 0.45 FPS (2226.3 ms)
```

Video (night time)

Car driving at night



```
A black car is driving down a street at night.  
17:36:27 | INFO | refresh rate: 0.47 FPS (2127.4 ms)  
[gststreamer] gststreamer nicesink0 taglist, encoder=(string)  
uint)163, maximum-bitrate=(uint)4096000, nominal-bitrat  
A black car is driving down a street at night.  
17:36:29 | INFO | refresh rate: 0.45 FPS (2221.8 ms)  
A black car is driving down a street at night.[gststreamer]  
st, video-codec=(string)"H.264\ \ (High\ Profile\)", min  
, maximum-bitrate=(uint)6033600, bitrate=(uint)1533484;
```

Video night time)

Parked Bus



a white bus is parked on the side of the road
17:36:23 | INFO | refresh rate: 0.45 FPS (2218.7 ms)
a white bus is parked on the side of the road
17:36:25 | INFO | refresh rate: 0.46 FPS (2157.1 ms)

Video (night time)

2 People riding a bike on the street at night.

(0.45 FPS)

```
17:36:46 | INFO | refresh rate: 0.45 FPS (2217.5 ms)
2 people riding a bike on the street at night.
17:36:49 | INFO | refresh rate: 0.43 FPS (2322.6 ms)
2 people riding a bike on the street at night.
17:36:51 | INFO | refresh rate: 0.45 FPS (2205.6 ms)
2 people[gstreamer] gstreamer nicesink0 taglist, encoding=
(version=(uint)163, maximum-bitrate=(uint)4096000, nominal-bitrate=(uint)4096000,
are riding bikes on the street at night.
17:36:53 | INFO | refresh rate: 0.45 FPS (2205.4 ms)
2 people are riding bikes on the street at night.
17:36:55 | INFO | refresh rate: 0.45 FPS (2213.6 ms)
[gstreamer] gstDecoder -- end of stream (EOS)
```

Describe the image concisely.

A white bus is driving down a street at night.



Problem Statement

An **interpretable navigation policy** for intelligent **driving agents** trained using world model with explainable RL for autonomous driving in real world setting (at **action level**).

- Training agents on **World Models** (Internal WM) using RL/IL.
- Existing **explainable AI** (XAI): Vision Language Action Models (VLAMs) techniques does not explain the underlying “**action**” driving policy for agents behavior.
- Require **explainable RL** for interpretable features, policy and learning process.

Why?

- Current: Perception architectures based on explainable machine learning.
- Transition: LLMs (Large Language Models) → LWMs (Large World Models)
- Future: Cognitive Architectures based on explainable reinforcement learning.

Existing Efforts

Existing Efforts

XAI based Autonomous Vehicles

An autonomous vehicle provides a natural and intelligible **explanation** of its **real-time decision**.

AVs must also **explain** their AI-guided decision-making process.

Need to interpret the natural language based explanations for **actions**.

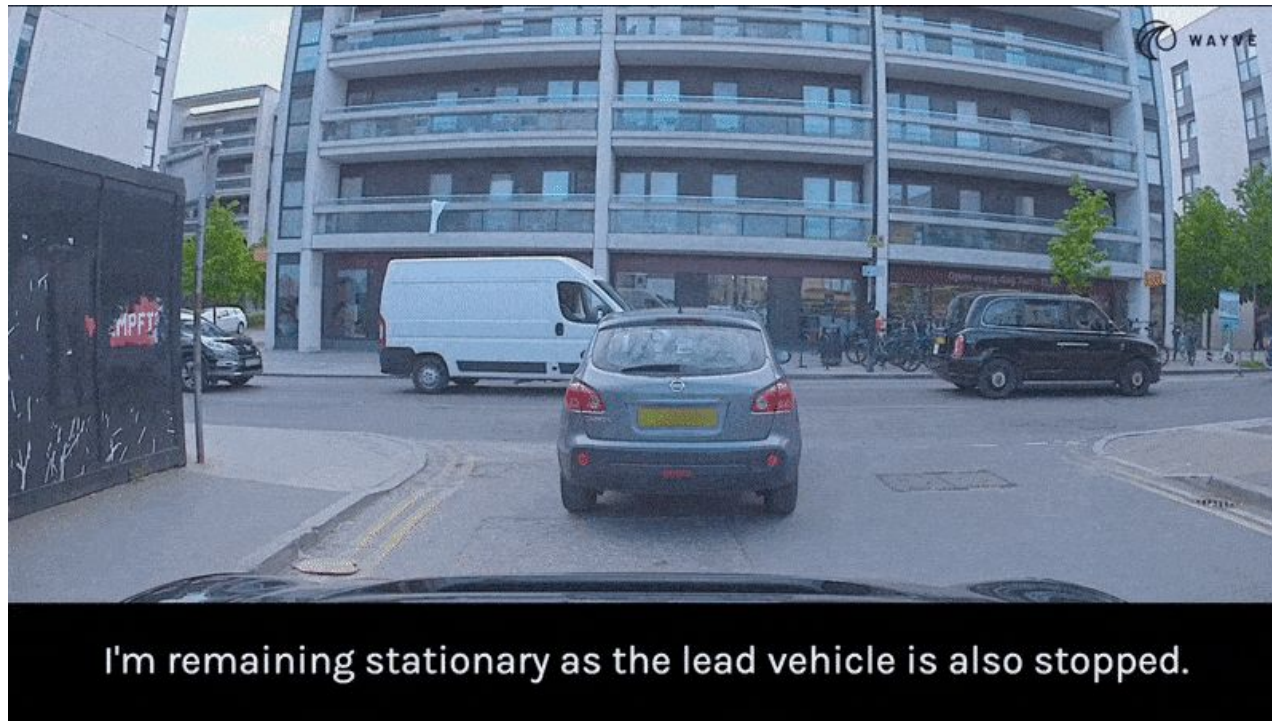


Ref: [Autonomous concept car smart vision EQ fortwo](#)

Explanation using VLAM

Objective

To build an agent to provide **reasoning** capabilities to an agent's **actions**.

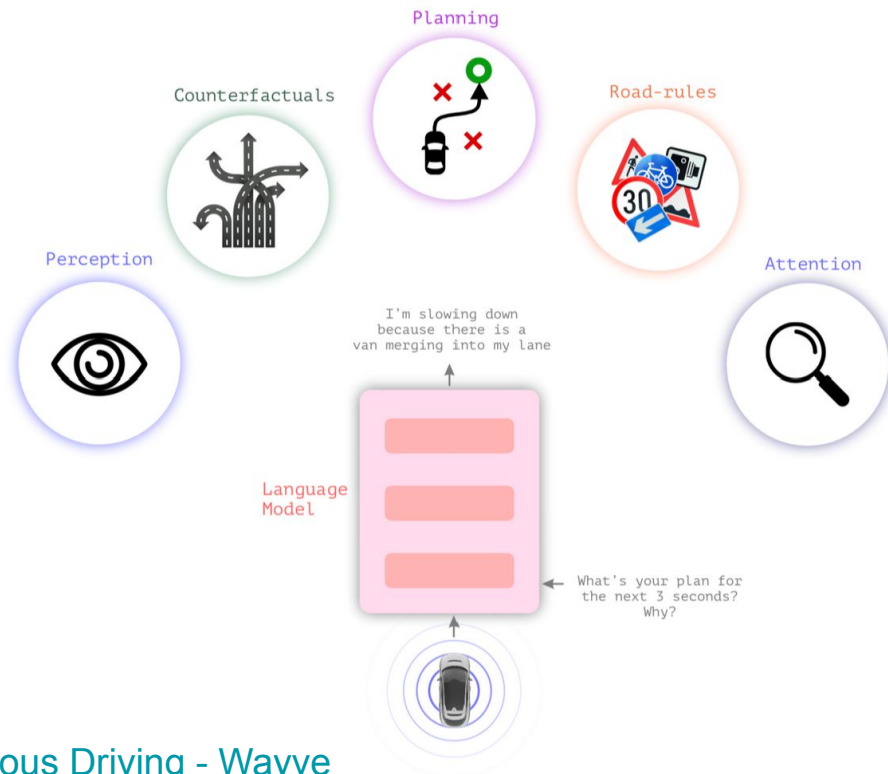


LINGO-1: Exploring Natural Language for Autonomous Driving

LINGO-1, an open-loop driving model, on various vision and language data sources perform visual question answering (VQA) on tasks such as **perception, counterfactuals, planning, reasoning and attention.**

Prompt LINGO-1 with questions regarding scene understanding and reasoning about the primary causal factors in the scene affecting the driving decision.

LINGO-1 can provide a description of the driving actions and reasoning.



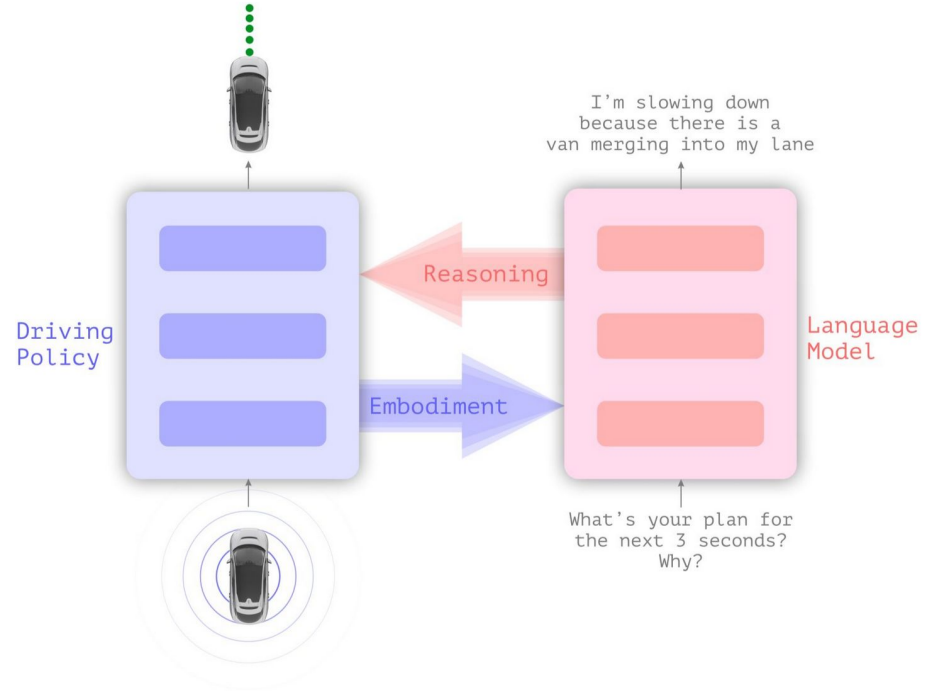
LINGO-1: Limitations

Need to improve driving performance through better planning and reasoning.

Two factors affect driving performance

- Ability of the language model to **accurately interpret scenes.**
- Model to translate reasoning into effective **low-level planning.**

Require interpretable driving policy.



LingoQA

Still the explanation by the agent about the action is a simple natural language explanation.

Driving Reasoning



Q: Should you merge into the lane to your left?

A: No, I should not merge into the lane to my left as it is reserved for buses and cyclists.

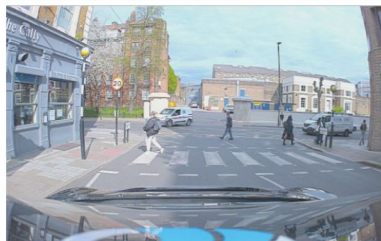
Object Recognition



Q: What objects can you see?

A: I can see a cyclist on the left, a bus on the left, a truck in the right lane, a car in the right lane.

Action Justification



Q: What is your next action and why?

A: Slowing down, pedestrians crossing the road.

Scene Description



Q: Please describe the scene ahead.

A: I am driving on a two-lane road in London, maintaining a safe distance from a pedestrian.

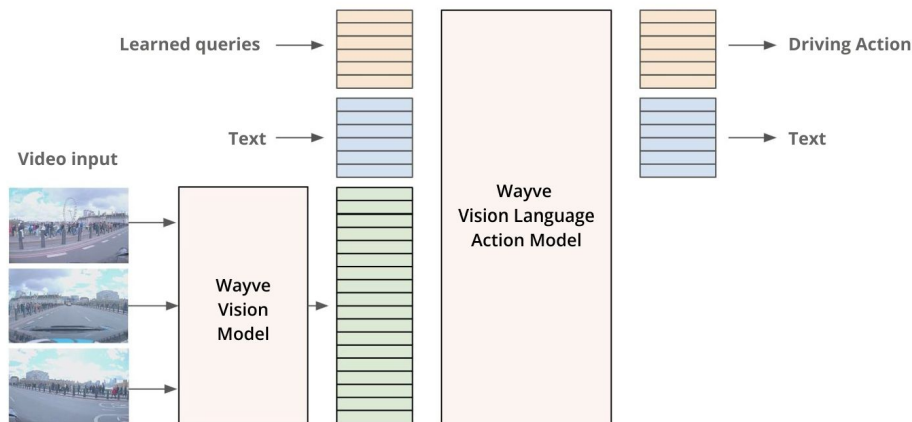
LingoQA is a benchmark for Video Question Answering in autonomous driving. Baseline vision-language model to answer questions related to driving reasoning, object recognition, action justification, and scene description.

LINGO-2: Driving with Natural Language

A driving model that links **vision**, **language**, and **action** to explain and determine driving behavior, opening up a new dimension of **control** and customization for an autonomous driving experience.

LINGO-2 is the first closed-loop vision-language-action driving model (VLAM) tested on public roads.

How to interpret these explanations?



Solutions

Why XRL? Explainable reinforcement learning is an emerging subfield of explainable ML.

The goal of XRL is to elucidate the **decision-making** process of reinforcement learning (RL) agents in **sequential** decision-making settings.

To understand: What the agents will do and why. A novel **taxonomy** for organizing the XRL.

Three high-level categories:

- ***Feature Importance,***
- ***Learning process and Markov decision process, (LPM) and***
- ***Policy-level (PL)***

Why? RL is combined with the *generalization and representational power of deep neural networks, which is often required to achieve the desired performance on these tasks.*

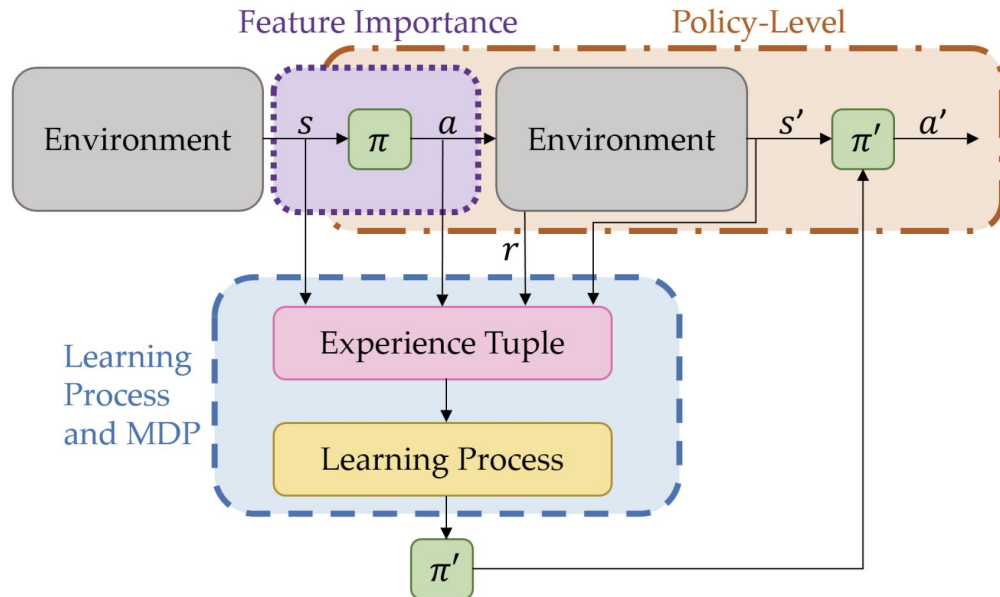
Solutions

XRL Taxonomy

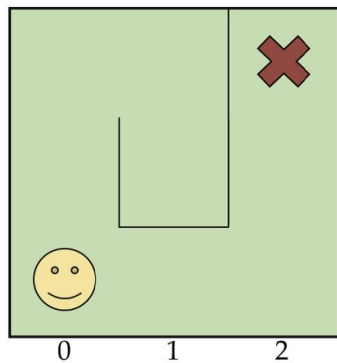
Proposed XRL taxonomy and its relationship to the RL problem.

Explanations may explain individual

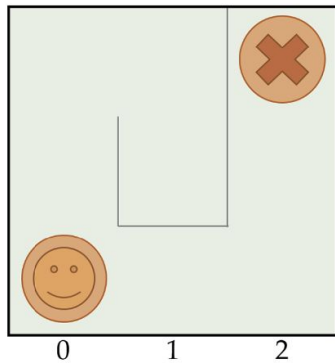
- **actions a** in terms of the input state s (**Feature Importance**),
- **experiences** from training (**Learning Process and MDP**), and
- Long-term behavior (**Policy-Level**)



Examples of different types of explanations



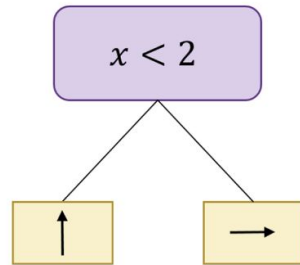
(a)



(b)

The agent
moves right to
reach the target
(X).

(c)



(d)

Given a **state** st from a domain (a), many *types of explanations* can be created, such as

- an **object saliency map** (b)
- a natural **language explanation** (c) and
- a decision **tree policy** (d)

Note that the explanations convey different information. For example, the natural language explanation (c) does not contain the counterfactual information of the decision tree policy (d).

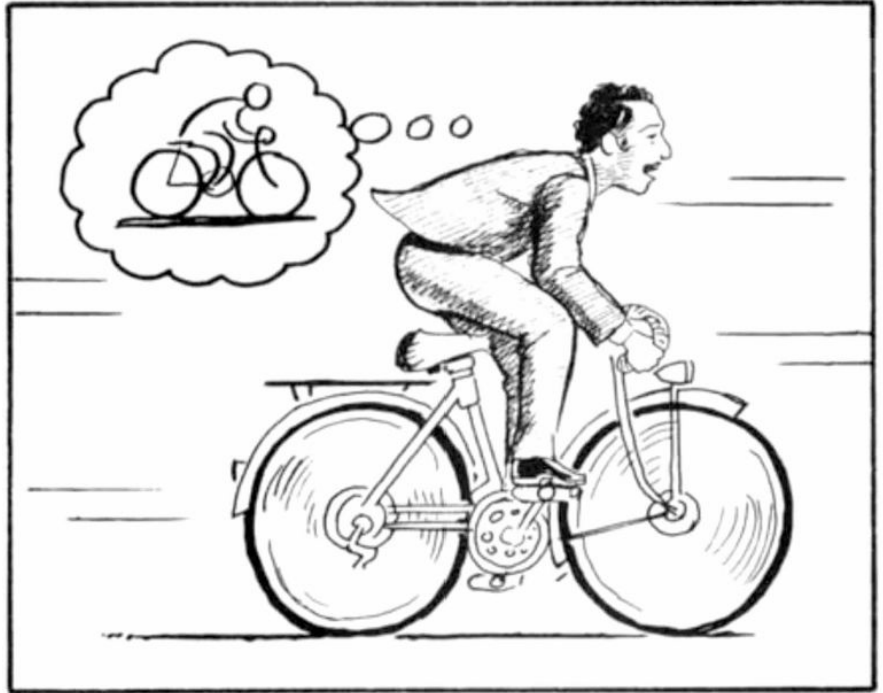
Implementation

World Models

Internal Model

“The image of the world around us, which we carry in our head, is just a **model**. Nobody in his head imagines all the world, government or country.

He has only selected **concepts**, and **relationships** between them, and uses those to **represent the real system**.”



World Models

Components of the Agent

Agent consists of three components that work closely together: **Vision (V)**, **Memory (M)**, and **Controller (c)**.

1. Agent has a **visual sensory** component that **compresses** what it sees into a small latent representation.
2. It also has a **memory** component that makes predictions about future codes based on historical information.
3. **Decision-making** component that decides what actions to take based only on the representations created by its vision and memory components.

At each time step, our agent receives an **observation** from the environment.

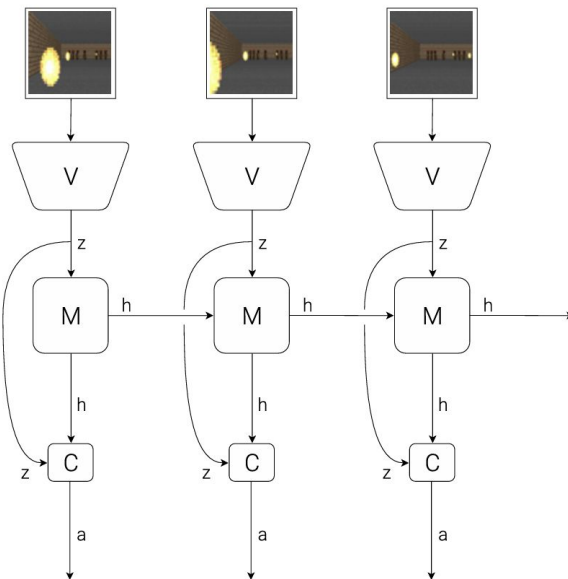
World Model

The **Vision Model (V)** encodes the high-dimensional observation into a low-dimensional latent vector.

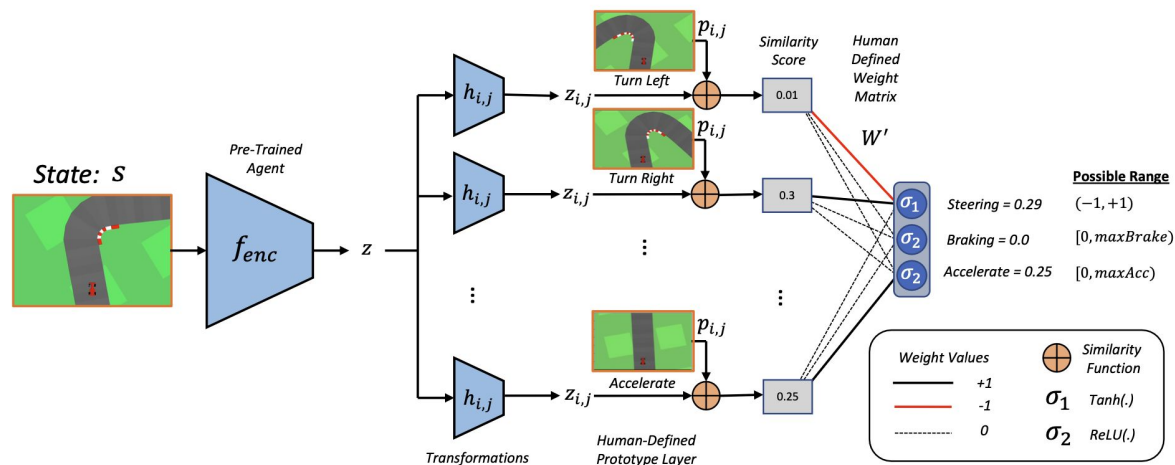
The **Memory RNN (M)** integrates the historical codes to create a representation that can predict future states.

A small **Controller (C)** uses the representations from both **V** and **M** to select good actions.

The agent performs **actions** that go back and affect the environment.



Prototype-Wrapper Network (PW-Net) - ICLR-2023



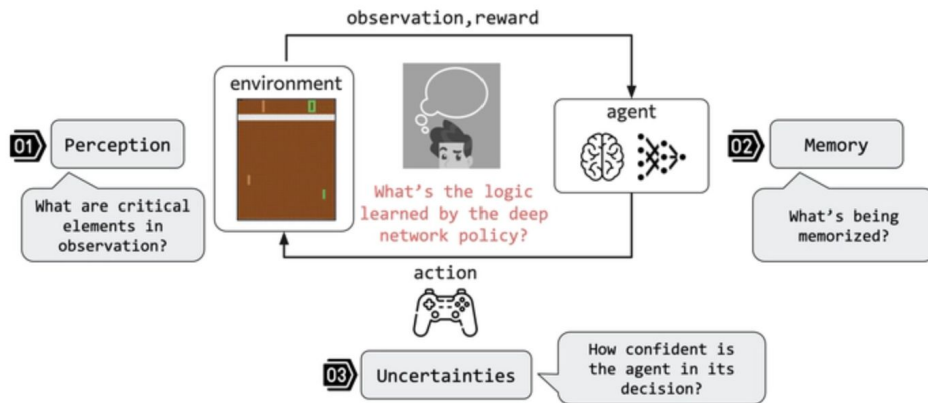
1. “Interpretable-by-design” deep RL agent which is forced to use human-friendly prototypes in its decisions, thus making its reasoning process clear. Inherently interpretable, well performing, deep RL algorithm.
2. (PW-Net), which creates a “wrapper” around the agent to make a new end-to-end interpretable policy that reasons with human-defined prototypes. PW-Net, converts black-box neural models into prototype-based agents for RL by forcing them to use human-understandable concepts, thus creating a new policy.
3. A car racing domain in which a simulated car must steer, brake, and accelerate.

XRL Agents from the lens of perception, memory, and uncertainties

Understanding causal reasoning behind decision making of RL agents is vital for enhancing trust and adaptability in real-world scenarios.

Several factors influence the decision-making process of these agents, including

- How they perceive the world,
- Utilize their memory in conjunction with perception, and
- Express uncertainties inherent in decision-making.



Ref: [XRL Agents from the lens of P,M,U. | Anurag Koul](#)

Objectives

- First, demo of a **VLM on an edge device with output results.**
- Second, baseline implementation of a simple world model based on RL.
- Third, reproduce a baseline paper on prototype bases agent for RL.
- Fourth, possibility of extending the research for explainability.
- Lastly, discuss XRL categorization and future of RL research.

References

1. [World Models](#) (2018)
2. [Towards Interpretable Deep RL with Human-Friendly Prototypes](#) (2023)
3. [The Alberta Plan for AI Research](#) (2023)
4. [LINGO-2: Driving with Natural Language - Wayve](#) (2024)
5. [LingoQA: Video Question Answering for Autonomous Driving](#)
6. [Explainable Reinforcement Learning: A Survey and Comparative Review](#) (2024)

World Models

Can agents learn inside of their own dreams?

DAVID HA Google Brain Tokyo, Japan	JÜRGEN SCHMIDHUBER NNAISENSE Swiss AI Lab, IDSIA (USI & SUPSI)
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March 27
2018

NIPS 2018
Paper

YouTube
Talk

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Abstract

- We explore building **generative neural network** models of popular **reinforcement learning** environments.
- Our **world model** can be trained quickly in an unsupervised manner to learn a **compressed** spatial and temporal **representation** of the environment.
- By using features extracted from the world model as inputs to an agent, we can **train** a very **compact and simple policy** that can solve the required task.
- We can even **train our agent** entirely **inside** of its own **dream environment** generated by its world model, and transfer this policy back into the actual environment.

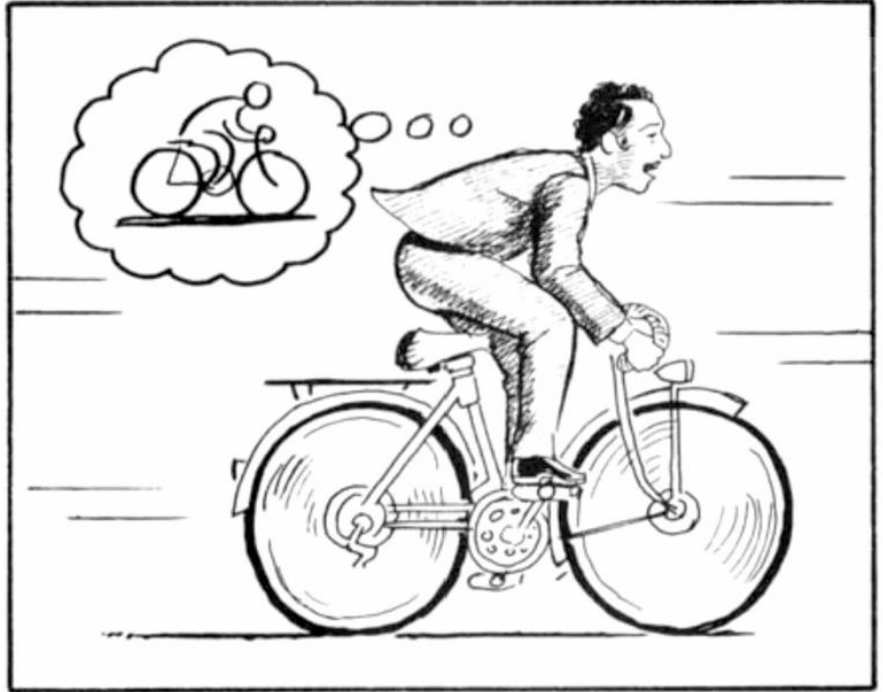
Introduction

Humans develop a **mental model** of the world based on perception with their senses. ***The decisions and actions we make are based on this internal model.***

Jay Wright Forrester, the father of system dynamics, described a mental model as:

*“The image of the world around us, which we carry in our head, is just a **model**. Nobody in his head imagines all the world, government or country. He has only selected **concepts**, and **relationships** between them, and uses those to **represent the real system**.”*

To handle the vast amount of information that flows through our daily lives, our brain learns an abstract representation of both spatial and temporal aspects of this information.



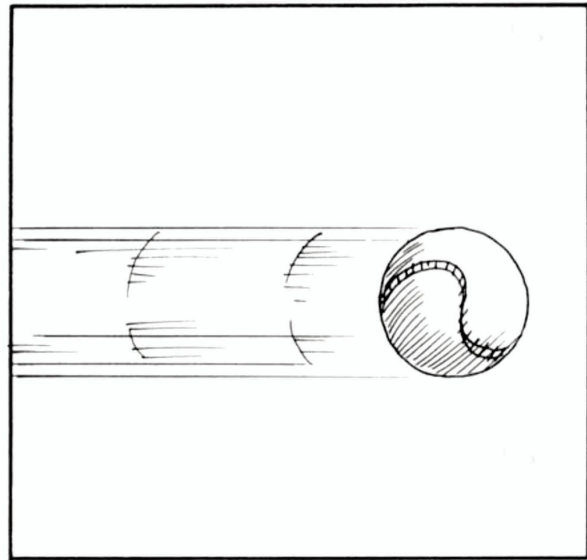
A World Model, from Scott McCloud's *Understanding Comics*

Baseball Example

Take baseball for example. A baseball batter has milliseconds to decide how they should swing the bat — shorter than the time it takes for visual signals from our eyes to reach our brain.

The reason we are able to hit a 100mph fastball is due to our ability to instinctively predict when and where the ball will go. For professional players, this all happens subconsciously. Their muscles reflexively swing the bat at the right time and location in line with their internal models' predictions.

They can quickly **act on their predictions of the future** without the need to consciously roll out possible future scenarios to form a plan.



Act on the predicted future state of the model.

Objective

An artificial agent also benefits from having a good representation of past and present states, and a good **predictive model** of the future.

To train a large neural network to tackle RL tasks, by **dividing the agent** into a large world model and a small controller model.

To learn a large agent's world model in an unsupervised manner, and then train the **smaller controller model** to learn to perform a task using this world model.

A small controller lets the training algorithm focus on the credit assignment problem on a small search space, while not sacrificing **capacity** and **expressiveness** via the larger world model.

Adjust a **temperature** parameter of internal world model to control the amount of uncertainty of the generated environments.

Agent Model

Agent consists of three components that work closely together: **Vision (V)**, **Memory (M)**, and **Controller (C)**.

1. Agent has a **visual sensory** component that **compresses** what it sees into a small latent representation.
2. It also has a **memory** component that makes predictions about future codes based on historical information.
3. **Decision-making** component that decides what actions to take based only on the representations created by its vision and memory components.

At each time step, our agent receives an **observation** from the environment.

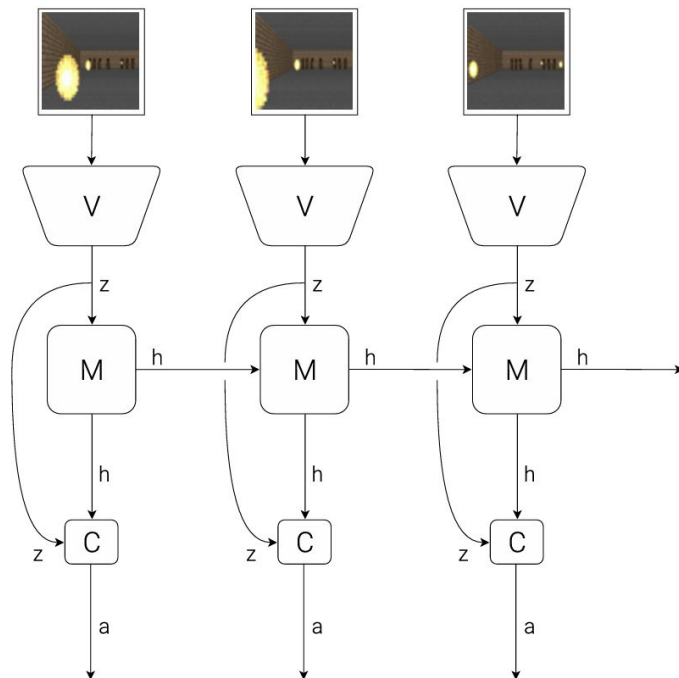
World Model

The **Vision Model (V)** encodes the high-dimensional observation into a low-dimensional latent vector.

The **Memory RNN (M)** integrates the historical codes to create a representation that can predict future states.

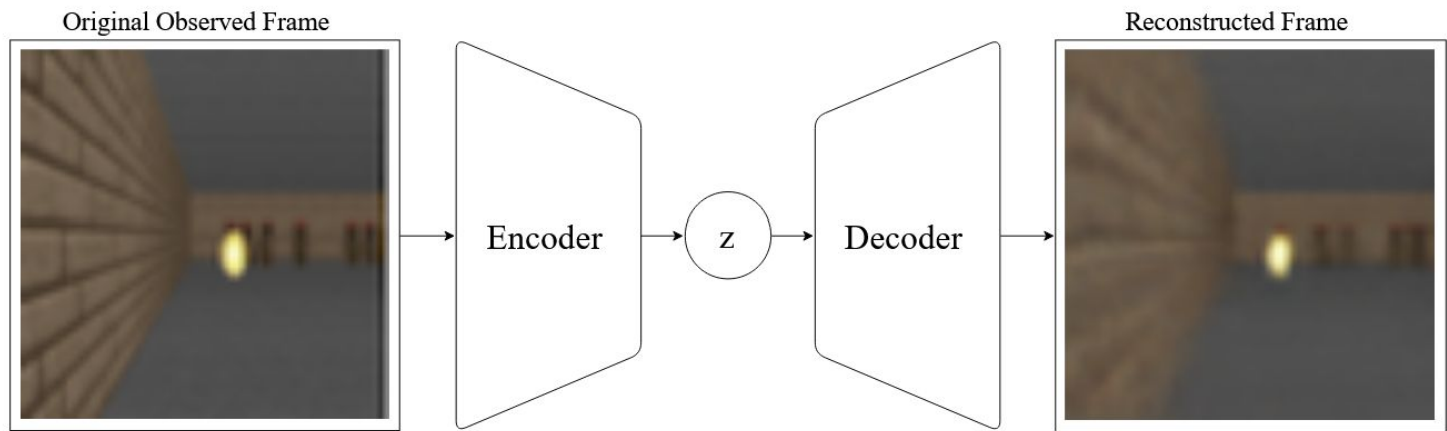
A small **Controller (C)** uses the representations from both **V** and **M** to select good actions.

The agent performs **actions** that go back and affect the environment.



VAE (V) Model

The environment provides our agent with a high dimensional input observation at each time step. This input is usually a 2D image frame that is part of a video sequence. The role of the V model is to learn an abstract, compressed representation of each observed input frame.



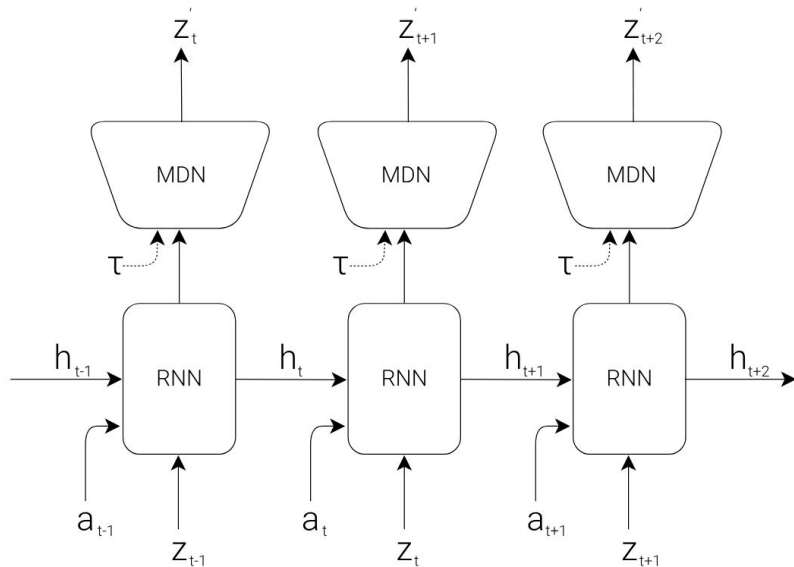
MDN-RNN (M) Model

Role of the V model to compress what the agent sees at each time frame.

Now, to compress what happens over time.
Role of the M model is to predict the future.

The M model serves as a predictive model of the future z vectors that V is expected to produce.

Because many complex environments are stochastic in nature, we train our RNN to output a probability density function $p(z)$ instead of a deterministic prediction of z .



RNN with a Mixture Density Network output layer. The MDN outputs the parameters of a mixture of Gaussian distribution used to sample a prediction of the next latent vector z .

a_t is the action taken at time t and h_t is the *hidden state* of the RNN at time t .

Controller (C) Model

The Controller (C) model is responsible for determining the **course of actions** to take in order to maximize the expected cumulative reward of the agent during a rollout of the environment.

In the experiments, C is as simple and small as possible, and trained separately from V and M, so that most of our agent's complexity resides in the world model (V and M).

C is a simple single layer linear model that maps **zt** and **ht** directly to action **at** at each time step:

$$\mathbf{a_t} = \mathbf{W_c}[\mathbf{z_t} \ \mathbf{h_t}] + \mathbf{b_c}$$ (This can be a complex non-linear function for motion planning)

In this linear model, $\mathbf{W_c}$ and $\mathbf{b_c}$ are the weight matrix and bias vector that maps the concatenated **input vector** **$[\mathbf{Z_t} \ \mathbf{H_t}]$** to the output action vector **$\mathbf{a_t}$** .

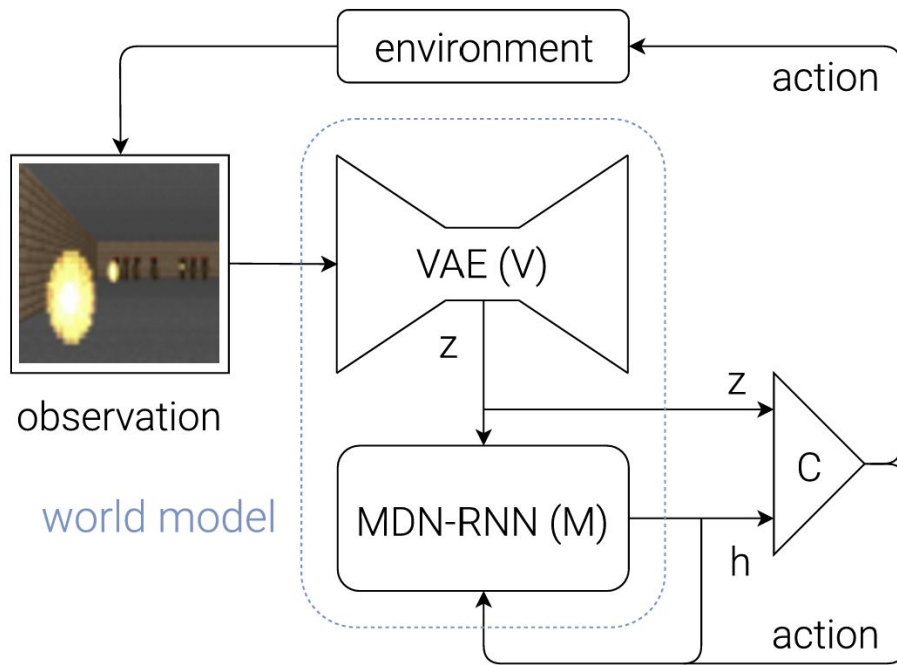
Full Model for the Agent

The raw observation is first processed by V at each time step t to produce z_t .

The input into C is this **latent vector** z_t concatenated with M 's **hidden state** h_t at each time step. (Mixture Density MDN)

C will then output an **action** vector at for **motor control**, and will affect the environment.

M will then take the **current** z_t and **action** a_t as an input to update its own hidden state to produce h_{t+1} to be used at time $t + 1$.



Car Racing : World Model for Feature Extraction

A predictive world model can help us extract useful representations of space and time.

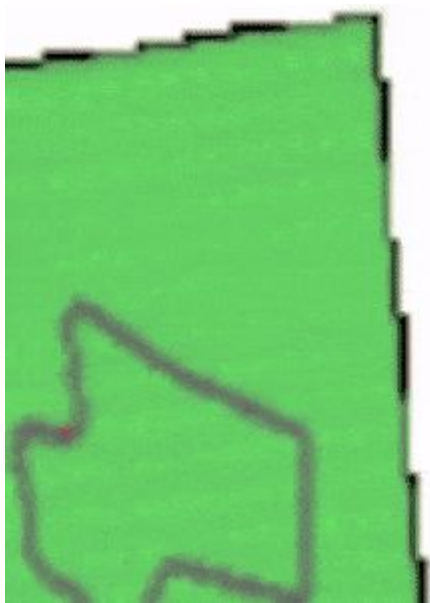
By using these features as inputs of a controller, we can **train a compact and minimal controller** to perform a continuous control task, such as **learning to drive from pixel inputs for a top-down car racing environment**.



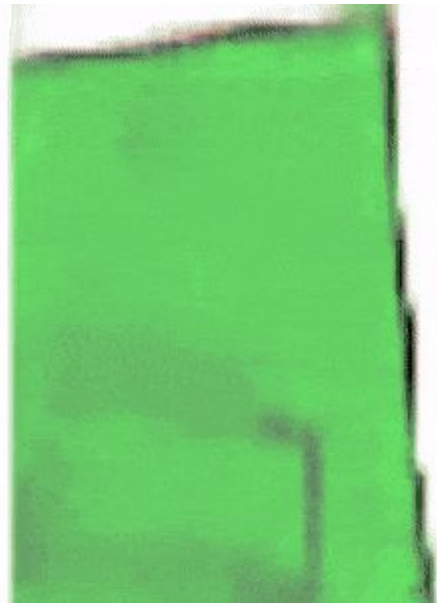
Observations and Latent Representations

Model trained on a car data environment frame by frame.

VAE to reconstruct each frame using z_t at each time step to visualize the quality of the information the agent actually sees during a rollout.



Actual observations from the environment.



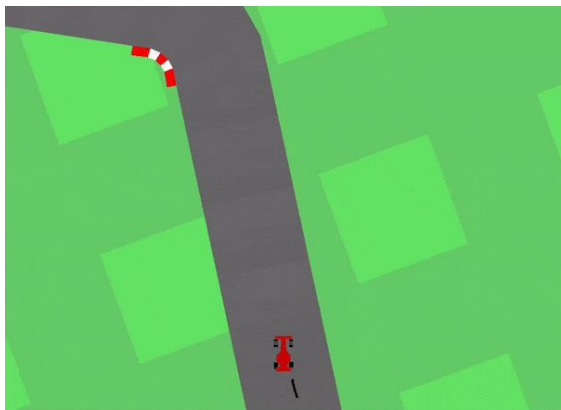
What gets **encoded** into z' .

Agent controls three continuous actions: steering left/right, acc, and brake.

Navigation Policy

Controller (C) access to only V C access to V and M (Full Model)

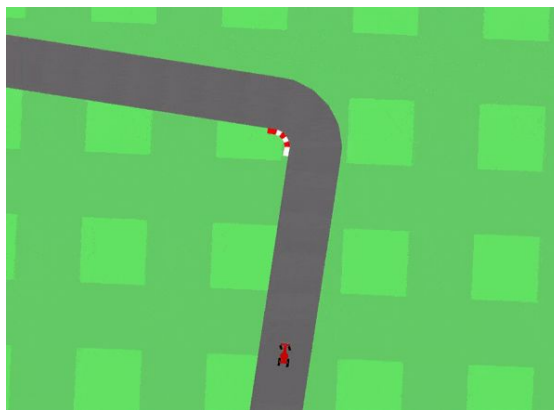
Wobbly navigation misses track



$$a_t = W_c z_t + b_c$$

Combining z_t and h_t provides good representation of observation.

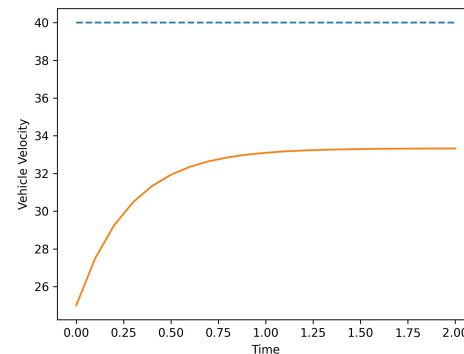
Better navigation **control**



$$a_t = W_c [z_t \ h_t] + b_c$$

Agents with (Stable) driving policy with V and M for C controller.

PID Control



Utilize real world concept space

Straight Road



Right Turn



Left Turn



Curve Road



Autoencoder

- Trained an autoencoder (AE) to learn latent representation of the external environment using camera images.
- To train the RL agent to learn driving policy without using waypoints.



Thanks